ICEBED V1.0

ICe volumE and Bedtopography Estimation from DEM

Instructions

ICe volumE and Bedtopography Estimation from DEM (ICEBED)

Overview

ICEBED is a software tool that helps estimate the ice thickness, ice volume, and ice bed topography of a glacier. This tool features a user-friendly GUI, enabling users to estimate the required outputs for one or multiple glaciers by following simple steps.

The ICEBED tool automates the GlabTop2_IITBversion ice thickness model (refer to Ramsankaran et al., 2018), an independently implemented version of the GlabTop2 model (Frey et al., 2014). This fully automated approach estimates spatially distributed ice thickness and bed topography reliable for glaciers lacking field measurements.

The differences between the GlabTop2_IITBversion and GlabTop2 models are shown in Table 1 (refer to Appendix of Thesis, Pandit, 2020).

Table 1: Differences between GlabTop2 and GlabTop2_IITB version model

S. No.	Environment/Processes	Model versions	
		GlabTop2	GlabTop2_IITB
1.	Programming environment	IDL	Matlab 2015a
2.	Model input	DEM and glacier mask	DEM, slope, and glacier mask
3.	Random pixel selection	“randomu”-IDL in-build function	“randperm” -Matlab in-build function
4.	Calculation of mean slope	mb_cellgradient.pro -user function	BufferProcessing.m -user function
5.	IDW interpolation	Basic	IDW interpolation with added distance weighting exponent

Outputs and Shape Factor Handling

ICEBED provides spatially distributed estimates of ice thickness and corresponding bed topography for the selected glacier(s). In addition, the tool generates (by default) a suite of ice thickness outputs corresponding to shape factor values ranging from 0.6 to 0.9 at increments of 0.01, resulting in 30 distinct thickness scenarios. The tool also computes, by default, an additional ice thickness dataset as the average of these 30 estimates, which is required for shape factor optimisation.

For applications adopting a standard shape factor, users may directly utilise the ice thickness estimates corresponding to a value of 0.8 or any other value. Alternatively, users seeking a more representative estimate may derive (manually) an optimised shape factor by following the methodology outlined in Ramsankaran et al. (2018), wherein the averaged ice thickness is used as an input to the optimisation framework. The optimised shape factor value obtained from this procedure enables the user to select the corresponding ice thickness estimate, which is considered as the final result when optimisation is applied.

How to Install?

To install this software, two files are required: the ICEBED.exe file and MATLAB Runtime software. You can access the MATLAB Runtime at https://in.mathworks.com/products/compiler/matlab-runtime.html and use MATLAB R2015a (8.5) based on the OS of your choice.

Once you select the .exe file, a window will appear (see below). Click Next and continue following the prompts.

The software will automatically check for MATLAB Runtime 2015a. If it is not present, you must install it manually by downloading it from the MATLAB website. Once installed, the software is ready for execution.

Steps to Estimate Ice Thickness Using a software tool ICEBED-ICe volumE and Bedtopography Estimation from DEM

1. **Required Inputs and formatting of Inputs for use:**

   a) Digital Elevation Model (DEM)
   b) Glacier Mask
   c) Glacuier Surface Slope

Note:

All the files must be in the .asc format. Convert the tif or jpeg images of DEM, MASK, and Slope files to .asc (ASCII) format first. These files should be in a single folder. If you use more than one glacier then each glacier should have separate folder containing respective DEM, Slope and Mask.

The input files must be clipped with the respective shapefile, keeping a buffer of 3 to 4 pixels outside the glacier boundary. All the files should maintain the same extent. The figure below (Figure 1) shows the three input files with similar rows and columns. If this is not correct, the software will show an error. Ensure that the NODATA values are assigned as -9999.

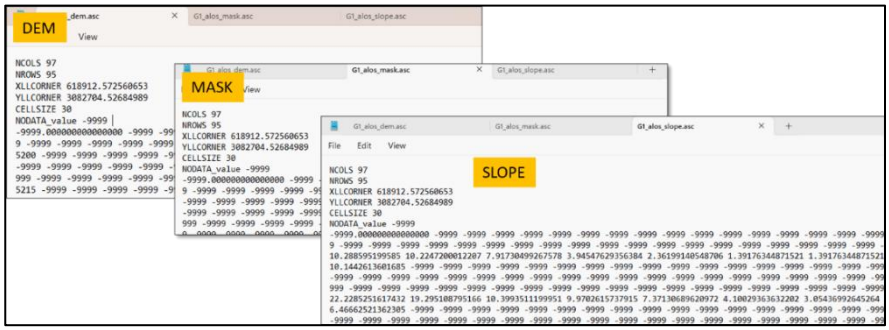


Figure 1: Illustration of the Input files in the .asc format

Once the above verification is done, remove the first to sixth rows (see Figure 2) in all the three input files.

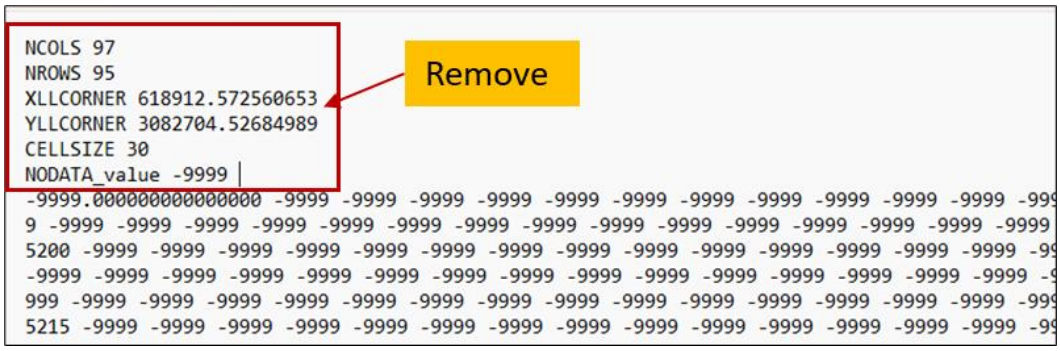


Figure 2: Details to be removed from the input files

2. Once the above process is done, the inputs are ready for estimating ice thickness.

3. Open the tool called as ICEBED- ICe volumE and Bedtopography Estimation from DEM, after installing the exe files. This tool follows the GlabTop2 IITB version wich is the improved version of GlabTop2 model approach (Ramsankaran et al, 2018). The display will be shown as in Figure 3.

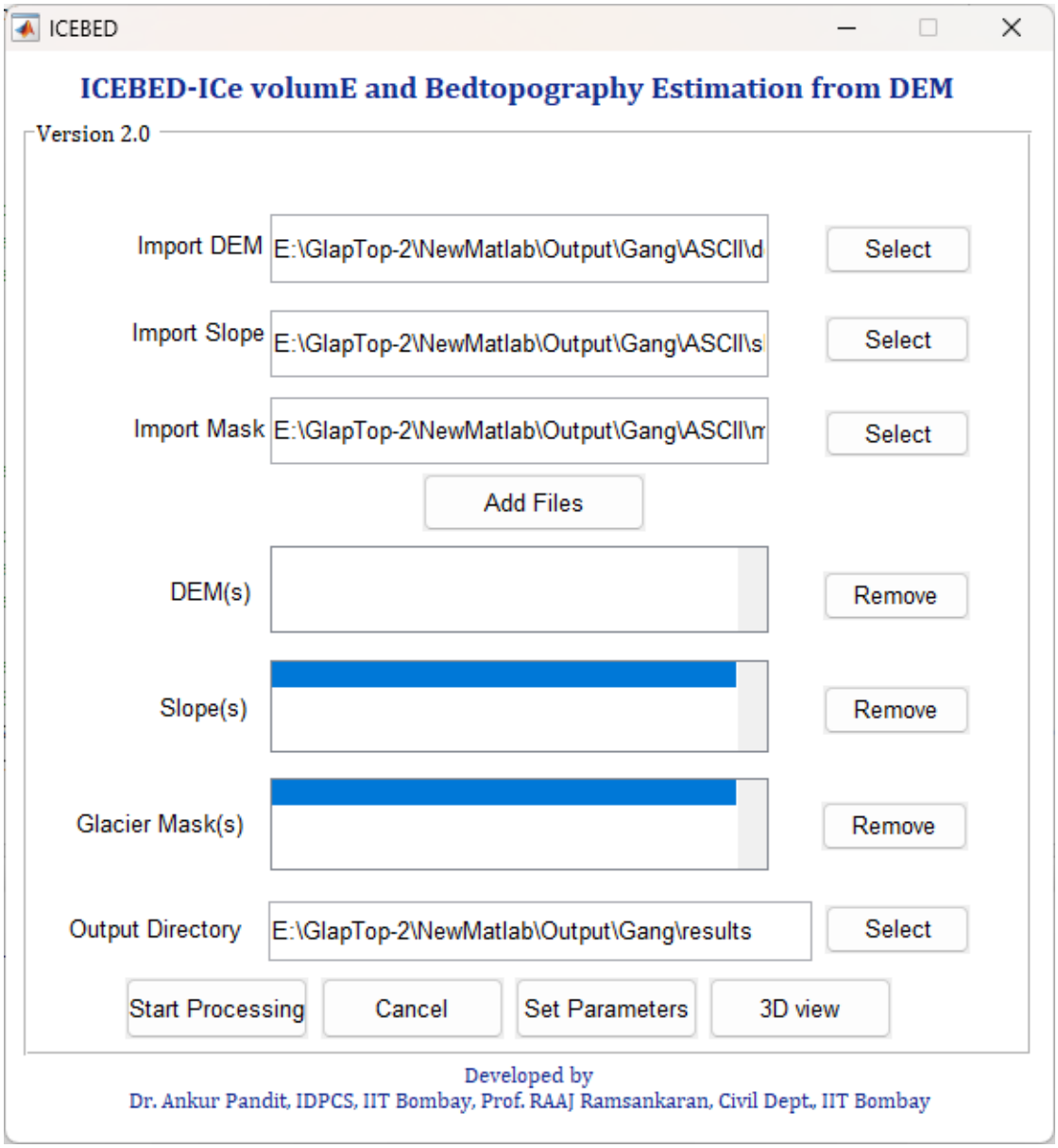


Figure 3: ICEBED tool with Graphical Use Interface (GUI).

4. After converting the input files, check the following:

a) In the "Import Data (DEM, Slope, and Mask)" section (see Figure 3), add all the appropriate inputs in the respective tabs.

i. Once you import the DEM, Slope, and Mask for a particular glacier (e.g., Hamtah), click the 'Add Files' button. This will add the inputs (i.e., DEM, Slope, and Mask) to the panel below.

ii. If you are interested in adding another glacier, simply import the DEM, Slope, and Mask as mentioned above and click the 'Add Files' button. This will add the inputs below the first glacier's (i.e., Hamtah Glacier) inputs.

iii. Repeat the same for multiple glaciers (see Figure 4). Depending on the processing power of your system, you can run the software for multiple glaciers. **Note:** Normally, 10-15 glaciers can be processed at once.

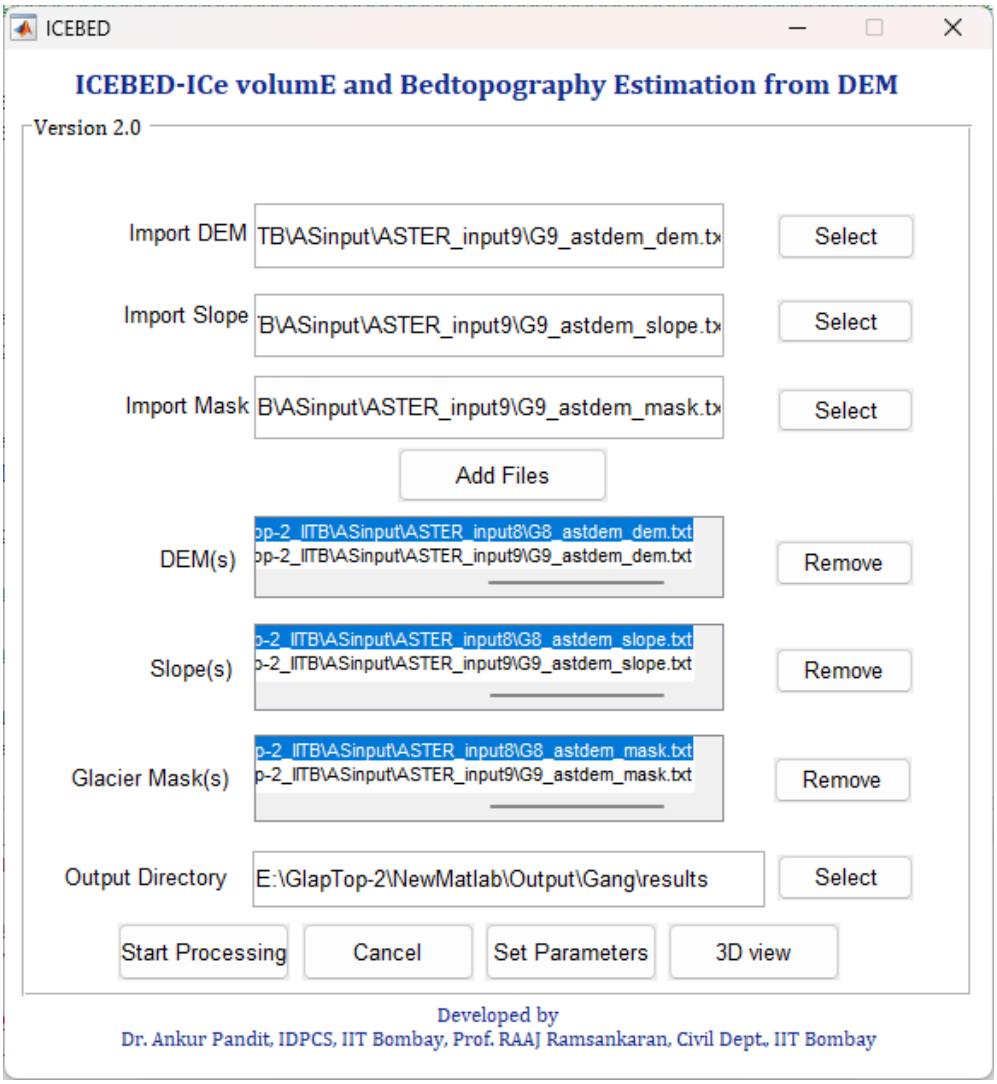


Figure 4: Illustration of adding more than one glacier in the input panels i.e. panels below 'Add Files' button.

b) Once all the necessary inputs are added, select the 'Add Files' button, and all the files will be added in the respective sections.

c) Next, to set the input parameters, select the 'Set Parameters' section.

d) The list of parameters will be displayed as shown in Figure 4. Here, only one input, i.e., ice density, needs to be supplied based on your evaluation. This parameter usually varies from 850 to 917 kg/m³.

Keep the remaining parameters (as shown in Figure 5) as they are. Based on the spatial resolution of the input DEM and Slope, assign the pixel size. Its unit is in meters. By default, it is set to '30m'. However, based on your need, the pixel value can be changed accordingly.

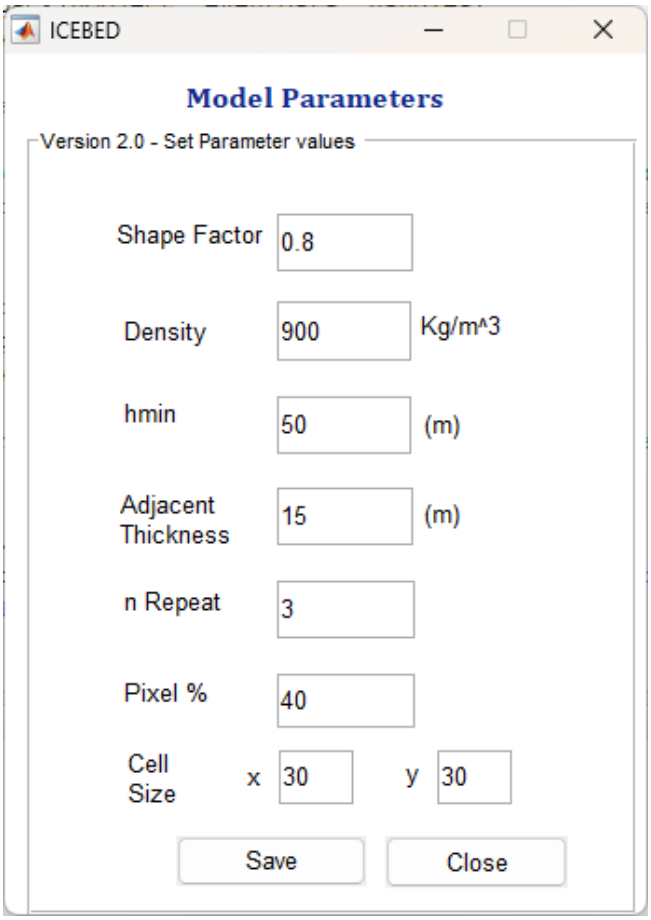


Figure 5: Set Parameters block in the ICEBED tool.

Note: The shape factor field is assigned a default value of 0.8, which represents the commonly adopted standard in glacier studies. However, irrespective of the value specified by the user, the tool internally computes ice thickness over a predefined range of shape factor values from 0.6 to 0.9 at increments of 0.01. Consequently, the model generates outputs corresponding to all shape factor values within this range. The final output directory therefore contains 30 individual folders (Figure 6), each representing results for a specific shape factor, with detailed outputs as illustrated in Figure 7. In addition, an averaged ice thickness dataset (computed as the mean of the 30 estimates) is also generated.

This design allows users to either directly adopt the standard shape factor value (0.8) or utilise the full range of outputs for subsequent optimisation, without the need for multiple model executions.

e) Once the parameters are set, provide the desired output folder location. **Note *(very imporant)*:** It is recommended not to use the input folder location.

f) Now that all the inputs and settings are done, select the 'Start Processing' button. The tool will automatically run and produce the following outputs (refer to Figure 6):

i. A folder containing mean ice thickness (i.e., the average of the spatially distributed ice thickness for shape factors ranging from 0.6 to 0.9 at every 0.01 interval).

ii. Other folders contain the ice thickness calculated for each shape factor value and corresponding bed topography estimates, metadata files, etc (refer Figure 7).

To know the inputs used and other details like shape factor value, mean ice thickness, and glacier volume, refer to the 'MetaDataFile' shown in Figure 8.

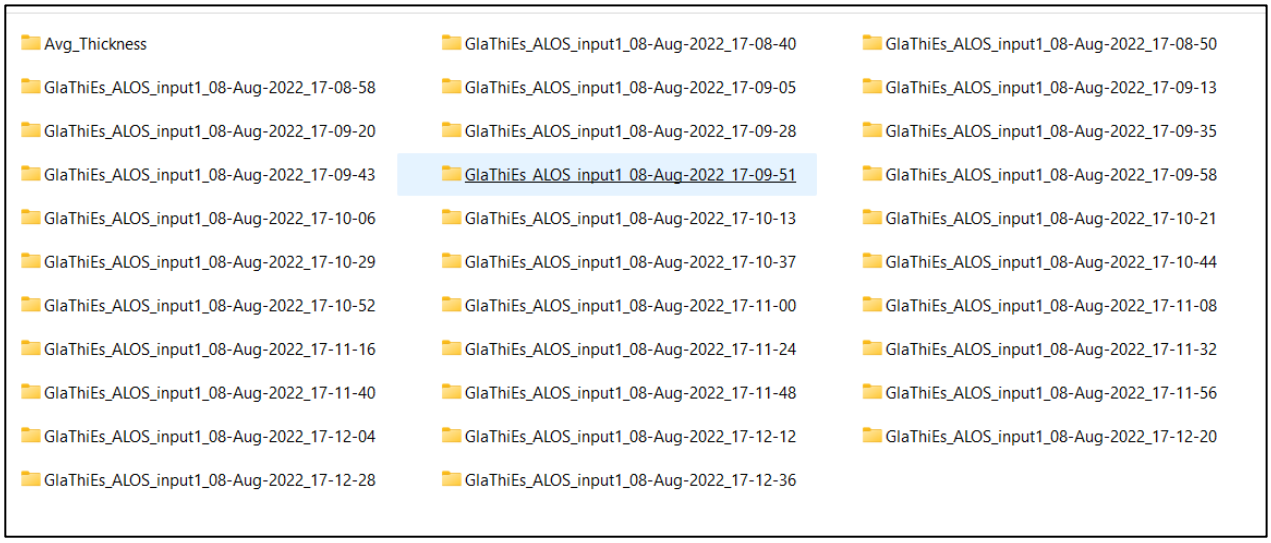


Figure 6: Snapshot of the output folders by ICEBED tool after the end of processing.

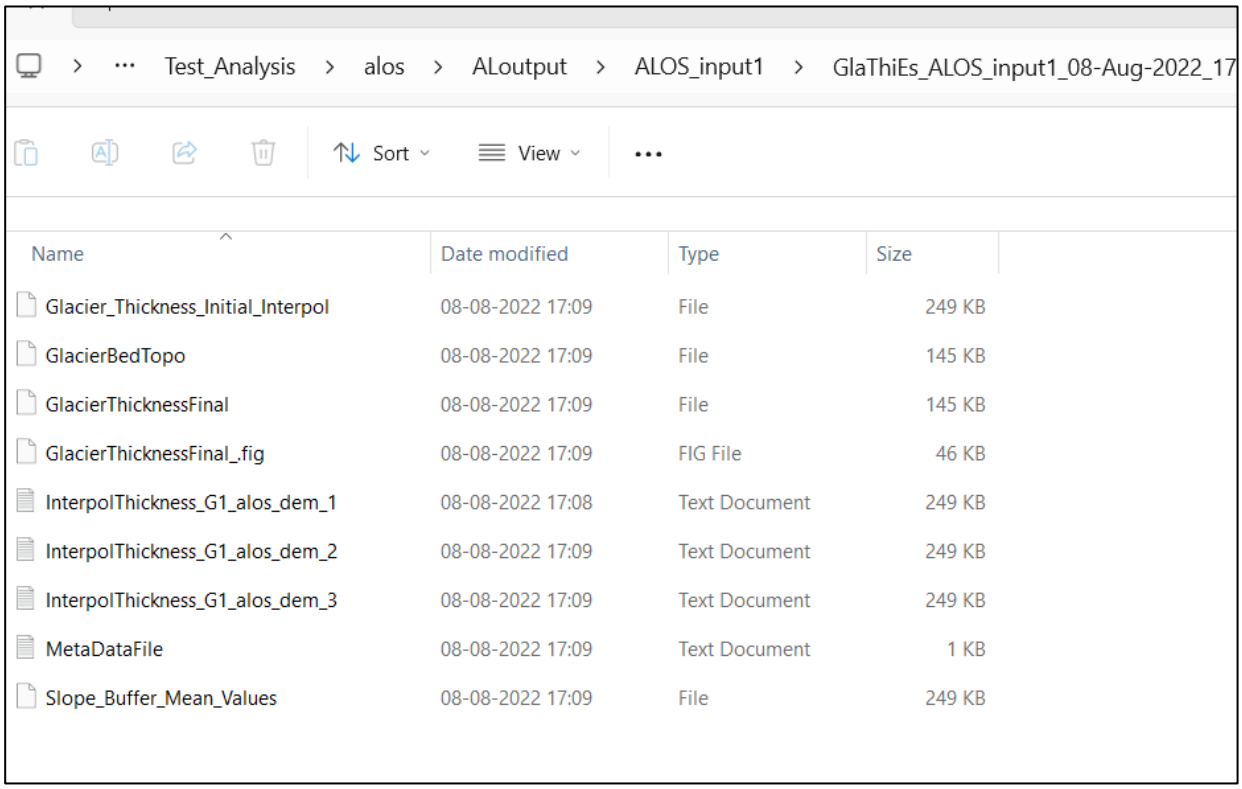
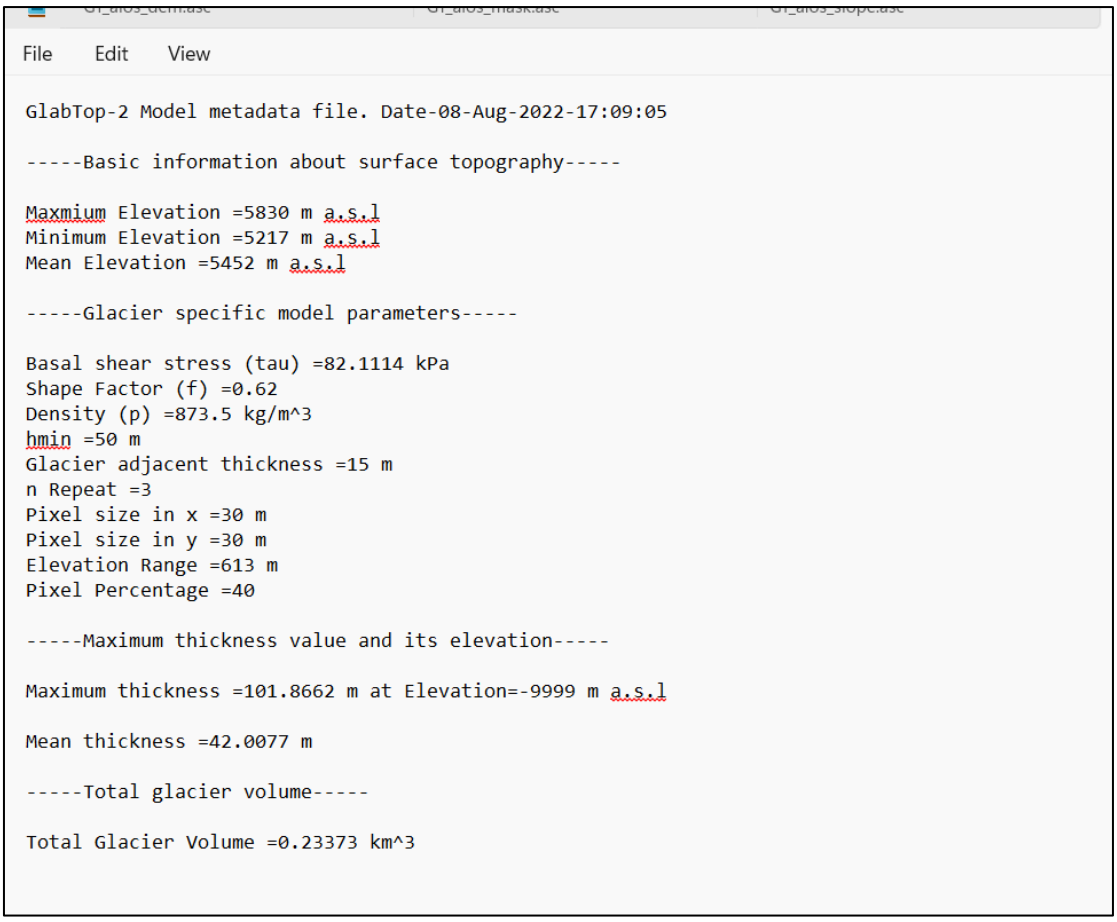


Figure 7: Snapshot of the folder showing the results present in each folder. It has Glacier ice thickness and bed topography



File Edit View

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GlabTop-2 Model metadata file. Date-08-Aug-2022-17:09:05

-----Basic information about surface topography-----

Maxmium Elevation =5830 m a.s.l
Minimum Elevation =5217 m a.s.l
Mean Elevation =5452 m a.s.l

-----Glacier specific model parameters-----

Basal shear stress (tau) =82.1114 kPa
Shape Factor (f) =0.62
Density (p) =873.5 kg/m^3
hmin =50 m
Glacier adjacent thickness =15 m
n Repeat =3
Pixel size in x =30 m
Pixel size in y =30 m
Elevation Range =613 m
Pixel Percentage =40

-----Maximum thickness value and its elevation-----

Maximum thickness =101.8662 m at Elevation=-9999 m a.s.l

Mean thickness =42.0077 m

-----Total glacier volume-----

Total Glacier Volume =0.23373 km^3
```

Figure 8: Illustration of MetaDataFile details available in each output folder (excluding Avg_Thickness)

5. Now we have output results such as:

1) Average ice thickness estimates.
2) Ice thickness, bed topography estimates, glacier volume, and metadata files for each shape factor values (0.6, 0.61…0.89, 0.9) in separate folders.

6. Based on the user interest, the above outputs can be used flexibly.

I. **If the user is fine with the ice thickness calculated using the widely adopted constant value of 0.8 or on any other constant value within the range 0.6 to 0.9**, they choose the appropriate ice thickness folder (refer to Figure 6) for preparation of ice thickness map of their study glaciers.

II. **If the user wants the ice thickness with optimized shape factor,** the following steps must be followed to obtain the shape factor-optimized ice thickness estimates:

   a. Firstly, we require the 'Average Thickness' file. Secondly, we need glacier cross-sections at every 100m elevation interval (one can either manually digitize the cross-sections of a glacier or obtain the existing cross-sections (in shapefile format) from the OGGM website available for global mountain glaciers, however, this file may require some revisions.

   b. Using the shape factor estimation formula (refer to Equation 3 of Ramsankaran et al. (2018)), determine the desired shape factor value. For example, if you get

0.68 as the shape factor value for your Glacier, choose the folder with ice thickness estimated for the shape factor value of 0.68.

To know the entire process in detail, refer to the Section 5.3 of Ramsankaran et al. (2018).

References:

Pandit, A., 2018. Glacier ice thickness modelling and identification of potential sites for future glacial lakes in the western Himalaya. PhD Thesis, Indian Institute of Technology Bombay, Mumbai, India.

Ramsankaran, RAAJ., Pandit, A., Azam, M.F., 2018. Spatially distributed ice-thickness modelling for Chhota Shigri Glacier in western Himalayas, India. Int. J. Remote Sens. 39, 3320–3343. doi:10.1080/01431161.2018.1430912